

JANTSZ / 12 FT V2 DOME

Geometry + clearance audit

A mathematical record of the dome axes, member classes, node families, the failed first tenon layout, and the port-normal redesign clearance model.

DOME GEOMETRY	CENTERLINE GEOMETRY VERIFIED
OLD JOINT FIT	FIRST TENON LAYOUT DOES NOT FIT

What this means

The dome mathematics are intact. The supplied tenon dimensions and former 3 in hub comparison envelope do not fit together: every tested pair of untrimmed tenon volumes intersects on the installed H4, H5, and H6 axes. That result applies to the old layout only; it does not invalidate the dome geometry and it is not a load test.

Do not order material, lay out joints, cut stock, machine hubs, set jigs, assemble, occupy, or scale from this review PDF.

Verified / withheld

VERIFIED IN THIS AUDIT	WITHHELD UNTIL A FABRICATION RELEASE
Topology, node coordinates and axes; two node-center chord classes; node valence and endpoint incidence; old-layout interference; defined port-normal redesign clearance model.	Released finished or blank timber lengths; structural connection capacity; retention; mortise fit and tolerances; hub layup; CNC and jig data; loads, anchorage, foundation, openings, code and occupancy.

Key node-center dimensions

PARAMETER	AUDITED VALUE	MEANING
Sphere radius	72.000 in	Center to mathematical node sphere
Base diameter	144.000 in	Opposite base nodes; not outside timber
Peak height	72.000 in	Base node plane to apex node
Base	Regular decagon	10 coplanar boundary nodes / 10 straight chords

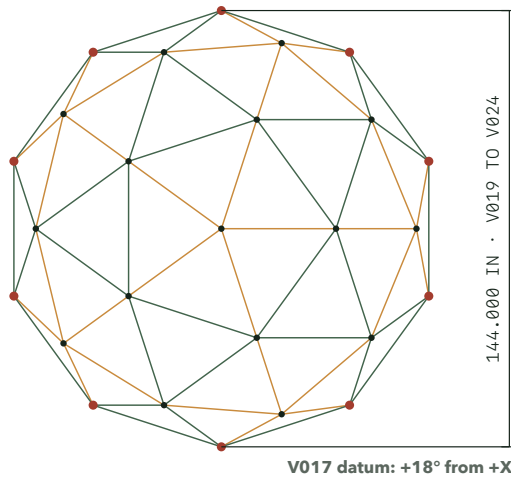
This downloadable document is a geometry and clearance audit—not a cut list, shop drawing, blueprint, permit set, structural design, or permission to fabricate.

SHEET 02 / CANONICAL MODEL

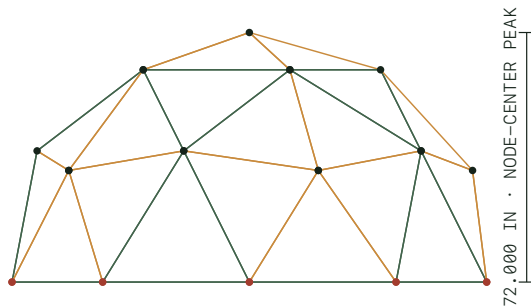
Plan and elevation

All lines terminate at mathematical node centers. Decimal precision records the computational model; it does not assert shop tolerance.

A-101 PLAN · NODE CENTERLINES



A-201 FRONT ELEVATION · NODE CENTERLINES



Coordinate datum

+Y is vertical. The base lies in the XZ plane at Y = 0. Base node V017 is +18° from +X and is located at (68.476069, 0.000000, 22.249224) in.

Member and node schedule

These counts describe the complete class-I frequency-2 icosahedral hemisphere generated by radial projection and an equatorial cut.

MODEL ITEM	COUNT	CHECK
Nodes V	26	10 base / 16 above base
Member axes E	65	130 directed endpoints
Triangular faces F	40	$V - E + F = 1$
Base boundary	10	Planar regular decagon

Two node-center chord classes

CLASS	COUNT	CHORD FACTOR	NODE-CENTER CHORD	ROLE
Short	30	0.5465330578 R	39.3503802 in	Shell
Long	35	0.6180339887 R	44.4984472 in	Shell + 10 base edges

Node valence families—not machining templates

FAMILY	COUNT	INCIDENT MEMBER AXES	LOCATION / LIMIT
H4	10	1 short + 3 long	Base; 5 positive / 5 reflected installed stars
H5	6	5 short	Apex + upper ring
H6	10	2 short + 4 long	Intermediate ring

Endpoint-pair groups

MEMBER	ENDPOINTS	COUNT	CENTERLINE
Short	H5–H6	20	39.3503802 in
Short	H4–H5	10	39.3503802 in
Long	H6–H6	15	44.4984472 in
Long	H4–H6	10	44.4984472 in
Long	H4–H4	10	44.4984472 in

Audit identities

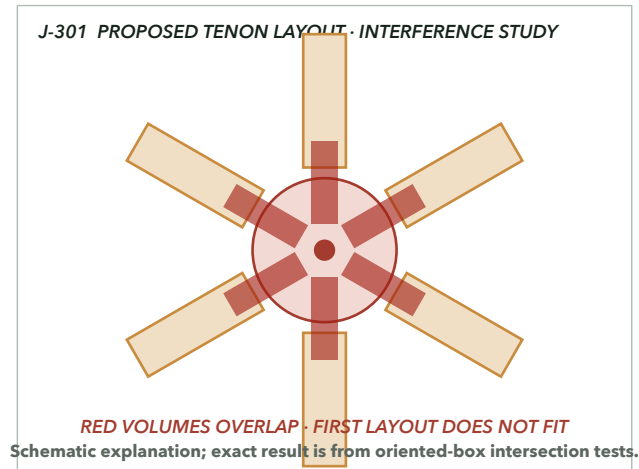
$$V - E + F = 26 - 65 + 40 = 1 \quad \cdot \quad 2E = 130 = 10(4) + 6(5) + 10(6)$$

Chord-axis equations: $L_{ij} = \|C_j - C_i\|$ and $u_{ij} = (C_j - C_i) / L_{ij}$. Short central angle 31.7174744° / tangent slope 15.8587372° ; long central angle 36.0000000° / tangent slope 18.0000000° .

Finished-length equations remain: $L_{\text{shoulder}} = L_{ij} - s_i - s_j$; $L_{\text{tip-to-tip}} = L_{\text{shoulder}} + t_i + t_j$. Page 06 evaluates the current CAD candidate at $S = 4.000$ in and $t = 1.250$ in, but no fabrication length is released.

What failed—and what did not

The proposed tenons overlap. The dome centerline geometry does not fail. The test below evaluates simplified, untrimmed oriented boxes at the stated common tangent-plane offset; it does not define a hub solid or connection capacity.



Withdrawn concept inputs used only for this interference test

INPUT	VALUE	BOUNDARY
Common tangent-plane apothem	1.500 in	Equal computational offset
Tenon test volume	1.5 L × 1.25 W × 0.5 T in	Untrimmed oriented box
Supplied envelope reference	3 in across / 2 in radial	Not a derived closed hub solid
Short / long setback	1.5593515 / 1.5771933 in	Computational tangent-face value

INSTALLED FAMILY	PAIR RESULT / HUB	INSTALLATIONS	TESTED ROLLS
H4	6 / 6 intersect	10	0° and 90°
H5	10 / 10 intersect	6	0° and 90°
H6	15 / 15 intersect	10	0° and 90°

Result

Every tested local pair intersects at the stated equal offset in both tested 90° rolls. This proves interference for those proposed test volumes only. No hub was clipped, mortise subtracted, clearance applied, cutter radius modeled, ligament or grain checked, wedge inserted, assembly path evaluated, or structural load applied.

Required before fabrication CAD resumes

Confirm actual dressed stock and moisture; species, grade, defects and grain; choose a hub and joint family; define retention and demountability; set ligament criteria, exterior/drainage/finish/adhesive, site loads, openings, anchorage, foundation, occupancy and tooling. Then complete exact solid collision, ligament, grain and assembly-access review, prototype/test evidence, structural review, and an explicit fabrication release.

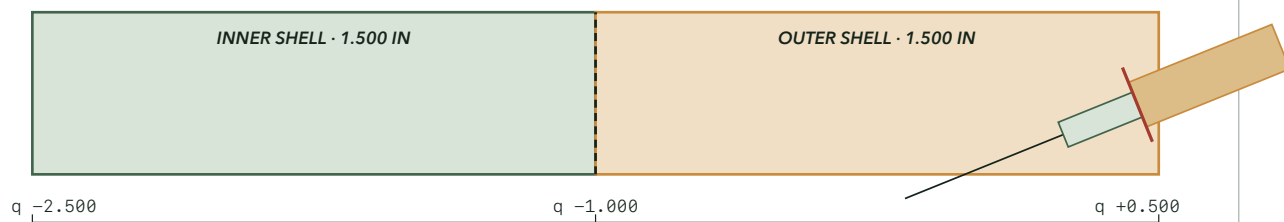
Redesign clearance model

This is the after-model for spatial review: each shoulder face is normal to its member axis at one fixed setback. It is not a selected or capacity-checked connection.

CLEARANCE AUDIT

SPATIAL CLEARANCE VERIFIED · STRUCTURAL REVIEW OPEN

J-302 PORT-NORMAL HUB ENVELOPE · CLEARANCE STUDY

SHOULDER FACE NORMAL TO MEMBER AXIS
TENON + OVERSIZED POCKET ENVELOPE

SCHEMATIC DATUM RELATIONSHIP · NOT A JOINT DETAIL

Defined digital envelope

INPUT	VALUE	AUDIT MEANING
Shoulder faces	Axis-normal at S = 4.000 in	One fixed axial datum per port
Exact convex footprints	H4 9.694 · H5 10.111 · H6 10.271 in	Maximum planar diameters
Tenon study	1.250 × 0.750 × 0.500 in	Integral volume study only
Oversized pocket	1.300 × 0.780 × 0.530 in	Collision envelope; not fit tolerance
Radial slab	q = -2.500 to +0.500 in	Relative to node radial datum
Two-shell division	q = -1.000 in	1.500 + 1.500 in shells
Member roll	width-radial · fixed	Fixed study condition
H4 base condition	1.500 in below datum	0.750 in rim; recess / raised datum required

Computed clearance record

CHECK	RESULT	MINIMUM SEPARATION
Oversized pocket pairs	0 of 24,570 collide	1.396637 in
Full 1.5 × 1.5 member pairs	0 of 24,570 collide	1.537727 in
Continuous-roll pocket bound	0 of 270 collide	1.380331 in
Split through every pocket	All installed pockets cross q = -1	0.347980 in each side
Full shoulder / other faces	Every 1.5 × 1.5 shoulder fits	1.011118 in
Key relief / center bore	Clear non-own pocket envelopes	2.221 / 1.738 in

Dimensional study boundary

The values below are consequences of the current CAD candidate. They make the model auditable, but they are not permission to cut timber or machine a connection.

Candidate member extents—not released cuts

CLASS	CENTERLINE	SHOULDER SPAN	MODELED TIP-TO-TIP
Short · 30	39.350380 in	31.350380 in	33.850380 in
Long · 35	44.498447 in	36.498447 in	38.998447 in

Connection boundary

Under TFEC 1-2019's standard wood-peg geometry, a pegged mortise-and-tenon cannot fit the modeled 1.500 × 1.500 in dressed stock. Any capture, cross-key, or clamp geometry is spatial exploration only. Retention, layup, grain, adhesive, fit, assembly, moisture behavior, loads, testing, and engineering remain open.

Reference: TFEC 1-2019 Standard for Design of Timber Frame Structures. Project-specific engineering remains required.

The PDF records dome geometry plus redesign clearance. It is not a cut list, shop drawing, blueprint, structural design, permit set, or fabrication release.

Audit trace

Geometry and old-layout sources: lib/geodesic.ts and lib/joinery.ts. Redesign values and metrics: REDESIGN_STUDY in lib/spec.ts, serialized by scripts/export_pdf_data.mjs. This PDF has no approval or signature block.